

Terafoods (TF) POS — Design System

****Target:**** Indoor Bangladeshi restaurants, 8–20 tables, 5–6 waiters per shift. Tier 2 small dine-in and Tier 3 full-service. Owners, managers, waiters share the same device or hand it between them.

****Hardware:**** Budget Android phones. MediaTek Helio G35 / Snapdragon 4-series. 3–4 GB RAM, PowerVR or Mali low-end GPU. 720p screens. Indoor lighting — fluorescent, LED panel, or tungsten. One-handed thumb operation expected.

****Locale:**** bilingual EN / BN at runtime. Bengali script must remain legible at 12–15 px.

****Product positioning:**** professional operating tool for restaurants. Not a consumer food app. The system must read as competent at every customer tier without fragmenting into multiple palettes.

****Source of truth:**** [`tf_design_system.dart`](../widgets/tf_design_system.dart) + [`app_theme.dart`](app_theme.dart). Every screen in `lib/src/features/` is built from the `Tf` primitives below.

1. Design principles

These six principles outrank every token below. When a token rule conflicts with a principle, the principle wins.

****Restraint over decoration.**** The accent appears on roughly 5% of any screen — primary CTAs, selected states, focus rings, the FAB. The other 95% is near-monochrome. Content carries personality. Chrome does not.

****Glanceability over density.**** The operator scans, does not read. Every screen must answer its primary question in under two seconds. Information that fails this test does not belong.

****Trust through consistency.**** A button looks like a button on every screen. A negative number is red on every screen. A success state is green on every screen. Predictability is the entire premium feel.

****Performance is a design constraint.**** A 200 ms animation jank during rush is a wrong order. The Helio G35 sets the ceiling, not the Pixel. Every visual decision is validated against the target device before it ships.

****Honest empty states.**** A zero value and a no-data value look different. Never show 0% when the answer is "not enough data yet."

****Multi-user reality.**** This is not a single-operator app. Five or six waiters share the device or carry their own. Every destructive action is attributable, every transient state is recoverable, every screen survives being handed mid-task.

2. Color system (`PosColors`)

Foundation — fixed, never changes

Token	Hex	Use
---	---	---
`paper`	`#FAFADF`	Scaffold background. Barely-warm off-white.
`surface`	`#FFFFFF`	Cards, sheets, fields, inputs.
`surfaceSunk`	`#F4F4F1`	Wells, depressed areas, search field fills.
`ink`	`#0F1419`	Primary text, headings, dark surfaces.
`inkSoft`	`#2A2F36`	Secondary dark, body emphasis.
`muted`	`#6B7280`	Secondary text, helpers, section headers.
`mutedSoft`	`#9CA3AF`	Tertiary text, placeholders.
`mutedFaint`	`#C7CAD1`	Disabled text, very faint dividers.
`line`	`#E5E7EB`	Default 0.5 px borders, dividers.
`lineStrong`	`#D1D5DB`	Card outlines on `paper`, emphasized borders.

Accent — deep coral-red

Established, slightly luxurious, food-coded without kitsch. Appears only on primary CTAs, selected states, focus rings, FAB, and accent-tinted soft surfaces. Maximum one full-saturation accent element per visible viewport.

Token	Hex	Use
---	---	---
`accent`	`#B73E3E`	Primary buttons, FAB, selected chips, focus rings.
`accentDeep`	`#8F2E2E`	Pressed states of accent buttons.
`accentSoft`	`#F2D5D5`	Selected-state backgrounds (ink text on top).
`accentWash`	`#F9E8E8`	Subtle accent backgrounds, hover, info-tinted callouts.
`accentInk`	`#FFFFFF`	Text and icons on full-strength accent surfaces.

Contrast: `accent` on `paper` is 5.4:1 (passes AA for text, AAA for large text). `accentInk` on `accent` is 5.8:1 (passes AA).

Signal — functional only

Signal colors carry meaning. Never decoration. Never on a value of zero unless the zero is itself a positive confirmation.

Because the accent is red, `danger` shifts darker to remain distinct. Side-by-side, accent reads warm and grounded; danger reads as alarm.

Token	Hex	Use
---	---	---
`success`	`#15803D`	Confirmed, healthy, positive deltas.
`successSoft`	`#DCFCE7`	Success-tinted surfaces.
`warning`	`#B45309`	Caution, low stock, attention needed.
`warningSoft`	`#FEF3C7`	Warning-tinted surfaces.
`danger`	`#7F1D1D`	Destructive, critical, negative deltas.
`dangerSoft`	`#FEE2E2`	Destructive-tinted surfaces.
`urgent`	`#9A3412`	Late orders, expedite signals.
`urgentSoft`	`#FFEDD5`	Urgent-tinted surfaces.

Data viz — five-color series

Calibrated to coexist without clashing with accent or signal.

Token	Hex
---	---
`data1`	`#1E3A5F` (deep blue)
`data2`	`#15803D` (matches `success`)
`data3`	`#7C3AED` (plum)
`data4`	`#92400E` (umber)
`data5`	`#0E7490` (deep teal)

`accent` is deliberately not in the data-viz set. Reserved for UI.

Color rules

- Hex literals are forbidden outside `app\_theme.dart`. Always `PosColors.\*`.
- `accent` never carries signal meaning. Negative delta uses `danger`. Positive delta uses `success`. Warning uses `warning`.
- Signal colors appear only when the underlying value is in that state. `urgent` is applied only when count > 0. At zero, render in `muted` text on `surface`.
- Maximum one full-saturation accent element per viewport. Selection states use `accentSoft` background with `ink` text.
- Soft surfaces (`successSoft`, `warningSoft`, `dangerSoft`, `urgentSoft`, `accentSoft`, `accentWash`) pair with `ink` or matching full-strength color, never with `muted`.
- Negative deltas use `dangerSoft` background with `danger` text. Positive deltas use `successSoft` with `success`. Neutral deltas use `surface` with `muted`.

- Third-party brand colors (bKash pink, Nagad orange) are not part of `PosColors`. If a screen needs one, scope it locally.

3. Typography

Families

- English → `Inter`
- Bangla → `Hind Siliguri`

Both have full numeric variants with tabular figures and render cleanly at small sizes on low-DPI screens. Always resolve via `tfFontFamily(context)`. String literals forbidden in feature code.

Weights

Weight	Use
---	---
`w400`	Body, helpers, descriptions, captions.
`w500`	Headings, buttons, prominent labels, numerals, emphasis.

`w600` and heavier are banned. Bangla conjuncts render unevenly at heavier weights on low-DPI screens. Hierarchy comes from size, color, surface contrast — never weight.

Tracking

Token	Value	Use
---	---	---
`tracking.tight`	`-0.02em`	Display sizes and `numericHero`.
`tracking.normal`	`0`	Everything else.
`tracking.wide`	`0.08em`	Uppercase eyebrow labels only.

The four ladders

Display, title, body, numeric. Each ladder serves a different purpose. Pick the ladder that matches the *role* of the text, not the size you want.

****Display ladder**** — hero numbers and screen titles, one per screen.

Style	Size	Weight	Tracking	Line height	Use
---	---	---	---	---	---
`displayHero`	44	w500	tight	1.05	The single largest number on a screen. One per viewport, max.

`displayLarge`	36	w500	tight	1.1	Secondary hero — order totals, large KPIs.
`displayMedium`	28	w500	tight	1.15	Featured analytics numbers.
`displaySmall`	24	w500	normal	1.2	Screen titles when not using `TfAppBar`.

****Title ladder**** — structural skeleton.

	Style	Size	Weight	Line height	Use
	---	---	---	---	---
`titleLarge`	18	w500	1.3		Card titles, dialog titles, list group headers.
`titleMedium`	16	w500	1.35		Sub-titles, prominent list items, alert primary.
`titleSmall`	14	w500	1.4		Dense list-row primary text, `sm` button labels.

Titles always pair with `ink`. If a title needs muted, it's a body label.

****Body ladder**** — connective tissue.

	Style	Size	Weight	Line height	Use
	---	---	---	---	---
`bodyLarge`	15	w400	1.5		Readable prose, onboarding copy.
`bodyMedium`	14	w400	1.45		Default body, list-row secondary.
`bodySmall`	12	w400	1.4		Helpers, captions, metadata.
`bodyTiny`	11	w400	1.35		Timestamps, badge counters. Use sparingly.

Line heights are generous — Bangla has vowel marks that need vertical room.

****Numeric ladder**** — every number, tabular figures mandatory.

	Style	Size	Weight	Tracking	Use
	---	---	---	---	---
`numericHero`	44	w500	tight		Dashboard hero number.
`numericLarge`	24	w500	normal		Featured KPI numbers.
`numericMedium`	18	w500	normal		Currency in list rows, KPI tile values.
`numericSmall`	14	w500	normal		Dense table cells, inline currency.
`numericTiny`	12	w500	normal		Badge counters, sparkline labels.

Never use a body style for a number. `bodyMedium` for "₹175" is wrong — use `numericMedium`. Body uses proportional figures; numeric uses tabular. Tabular aligns vertically; proportional doesn't.

Label ladder

	Style	Size	Weight	Tracking	Transform	Use
	---	---	---	---	---	---
`labelLarge`	14	w500	normal	none		Button labels (`lg`, `md`).

`labelMedium`	13	w500	normal	none	Chip labels, segmented controls.
`labelSmall`	12	w500	normal	none	`sm` button labels, status badges.
`labelEyebrow`	11	w500	wide	uppercase	Section eyebrows: "NEEDS YOUR EYE", "TOP MOVERS".

Eyebrow labels default to `muted`. Pass `color: PosColors.danger` for alert variants.

Hierarchy by context

****Dashboard hero card****

- Eyebrow: `labelEyebrow` muted
- Number: `numericHero` ink
- Delta: `numericSmall` in signal color
- Sparkline label: `bodySmall` muted

****KPI tile (4-up row)****

- Number: `numericMedium` ink
- Label: `bodySmall` muted

****Analytics card with sparkline****

- Eyebrow: `labelEyebrow` muted
- Number: `numericLarge` ink
- Delta + period: `numericSmall` + `bodySmall` muted

****Order list row****

- Order/table: `titleMedium` ink
- Status badge: `labelSmall` in `TfStatusBadge`
- Items summary: `bodyMedium` muted
- Total: `numericMedium` ink, right-aligned
- Time elapsed: `bodySmall` muted

****Menu item row (rush mode)****

- Name: `titleSmall` ink
- Description: `bodySmall` muted
- Price: `numericMedium` ink
- Stepper count: `titleMedium` ink

****Inventory row****

- Item name: `titleSmall` ink
- Category: `bodySmall` muted
- On-hand: `numericMedium` ink, unit in `bodySmall` muted
- Delta: `numericSmall` in signal color

****Section header****

- `labelEyebrow` muted — never `titleLarge`.

****Card title****

- `titleLarge` ink, with `sp3` below.

****Alert row****

- Primary: `titleMedium` ink
- Secondary: `bodySmall` muted
- Action: `labelSmall` in ghost button

****Empty state****

- Title: `titleMedium` ink centered
- Message: `bodyMedium` muted centered
- Action: primary button

****No-data state****

- Explanation: `bodyMedium` muted, single line. Pattern: "{Feature} appears once {condition}."

****Settings row****

- Label: `bodyLarge` ink
- Value: `bodyMedium` muted right-aligned

****Form field****

- Label above: `labelMedium` muted
- Input: `bodyLarge` ink
- Helper: `bodySmall` muted
- Error: `bodySmall` danger

Hierarchy by screen role

****Operational screens**** (orders, FOH entry, kitchen view) lean on title and body. Numeric appears for prices. Reading speed beats visual drama.

****Dashboard screens**** (Manage, Review) lead with display and numeric. One `numericHero` per screen, surrounded by `numericLarge`/`numericMedium`. Body recedes.

****Analytics screens**** mix all four ladders. Eyebrows organize, numeric carries data, body explains methodology in muted captions.

****Form screens**** dominated by body and label. Numbers only appear when typed.

Bilingual rules

- `TfText(string)` converts Western digits to Bengali numerals in BN locale.

- `TfMoney(amount)` formats with `₹` and locale-correct digits.
- BN strings render at +1 px line height vs EN. Handled internally by `TfText`.
- BN strings are 20–30% longer than EN. Never use fixed-width label containers.
- `TfTextPair(en: '...', bn: '...')` for parallel labels. Machine translation forbidden.
- Raw `Text(...)` banned. Always `TfText`.

Anti-patterns

- `bodyMedium` for currency in a list — use `numericMedium`.
- `fontSize: 19` because 18 felt small. Pick one and adjust spacing.
- Two `displayHero` on one screen — hierarchy has failed.
- A title in `muted` — not a title.
- Body at `w500` — body is `w400` always.
- Numbers in `bodyLarge` — numbers belong in the numeric ladder regardless of context.

4. Iconography (`TfIcon`)

Source

Lucide is the canonical icon library. Single stroke weight (2 px), geometric clarity, pairs cleanly with Inter, large enough to read on 720p. No other icon libraries permitted in feature code.

Imported via `lucide_icons` Flutter package. Wrapped through `TfIcon(LucideIcons.printer, size: TfIconSize.md)` — never use `Icon(LucideIcons.printer)` directly. The wrapper enforces sizing, color inheritance, and semantic labeling.

Sizes (`TfIconSize`)

Token	Value	Use
---	---	---
<code>`xs`</code>	14	Inline with body text, status badges.
<code>`sm`</code>	16	List-row decorators, inline accents.
<code>`md`</code>	20	Default. Button icons, list-row primary icons.
<code>`lg`</code>	24	Nav icons, FAB content, section headers.
<code>`xl`</code>	32	Empty states, large CTAs, onboarding.
<code>`xxl`</code>	48	Hero illustrations only. Reserve for first-run, empty hero states.

Color

Icons inherit parent text color by default. Override only when the icon carries independent signal meaning (e.g. a red trash icon in a delete confirmation; a green check on success).

Decorative icons in operational UI are banned. Every icon in `lib/src/features/\*\*` either:

- accompanies a label (button, list row, status badge), or
- represents an interactive control (toggle, action button) with a tooltip and semantic label.

A plain illustration in a card with no accompanying meaning fails the rule.

Canonical icons (`Tflcon.\*`)

These constants exist to prevent the same concept being represented by three different glyphs across the app. Use the constant, never the raw `Lucidelcons.\*` name.

Constant	Glyph	Use
--- --- ---		
`Tflcon.orders`	`clipboardList`	Orders nav, orders list.
`Tflcon.menu`	`utensilsCrossed`	Menu nav, menu items.
`Tflcon.home`	`house`	Home/dashboard nav.
`Tflcon.inventory`	`package`	Stock nav, inventory items.
`Tflcon.settings`	`settings2`	Settings nav, configuration.
`Tflcon.bell`	`bell`	Notifications, alerts.
`Tflcon.plus`	`plus`	New / add.
`Tflcon.minus`	`minus`	Decrement, remove.
`Tflcon.back`	`chevronLeft`	Back navigation.
`Tflcon.close`	`x`	Dismiss, cancel.
`Tflcon.check`	`check`	Confirm, completed.
`Tflcon.search`	`search`	Search.
`Tflcon.camera`	`camera`	AI scan, photo capture.
`Tflcon.printer`	`printer`	Print bill, print receipt.
`Tflcon.bluetooth`	`bluetooth`	Printer connection.
`Tflcon.wifi`	`wifi`	Network state.
`Tflcon.dot`	`circle` (filled)	Status dots.
`Tflcon.chevron`	`chevronRight`	List affordance, navigation.
`Tflcon.sparkle`	`sparkles`	AI features.
`Tflcon.flame`	`flame`	Urgent/late alerts.
`Tflcon.dineIn`	`armchair`	Dine-in order type.
`Tflcon.parcel`	`shoppingBag`	Parcel/takeaway.
`Tflcon.delivery`	`bike`	Delivery channel.
`Tflcon.table`	`square`	Table tiles.
`Tflcon.waiter`	`userRound`	Staff, waiter assignment.
`Tflcon.kitchen`	`chefHat`	Kitchen-related actions.
`Tflcon.cash`	`banknote`	Cash payment.
`Tflcon.card`	`creditCard`	Card payment.
`Tflcon.online`	`smartphone`	Online order channel.

New icons require updating this catalog and a one-line justification.

Rules

- Every icon-only button has a `tooltip` and an `accessibilityLabel`.
- Two-tone or filled icons forbidden — Lucide outlines only. `Tflcon.dot` uses Lucide's `circle` and sets `fill: PosColors.success` (or other) via the wrapper, but the icon itself remains a single stroke.
- Animated icons forbidden. The system uses a brief opacity fade on state change, not glyph morphing.
- Custom illustrations (empty states, onboarding) are allowed but must live in `assets/illustrations/` and follow the same stroke / palette discipline as Lucide.

5. Shape

Radii (`PosRadii`)

Token	Value	Use
---	---	---
`xs`	6	Chip pills, badges, tiny accents.
`sm`	10	Inputs, small buttons.
`md`	14	Cards, sheets, dialogs — workhorse.
`lg`	20	Featured cards, large surfaces.
`pill`	999	Fully rounded — selected chips, FAB, avatars.

Five values. No others. Exception: `circular(2)` for hairline accent stripes 2–4 px wide.

Borders

- All borders 0.5 px, using `line` or `lineStrong`.
- One border per surface. No nested outlines.
- Cards on `paper` use `lineStrong`. Cards inside cards use `line`.

6. Spacing (`PosSpacing`)

Token	Value	Use
---	---	---
`sp1`	4	Icon-to-label, badge padding.
`sp2`	8	Tight sibling spacing.
`sp3`	12	Default sibling spacing in cards.

`sp4`	16	Between card sections, screen edges.
`sp5`	24	Between major sections.
`sp6`	32	Around hero elements.
`sp7`	48	Breathing-room moments (rare on mobile).

Padding conventions

- Card content: `EdgeInsets.fromLTRB(sp4, sp3, sp4, sp3)`.
- Screen edges: `EdgeInsets.fromLTRB(sp4, sp4, sp4, sp4)`.
- Section gaps: `sp5` between major sections, `sp3` between siblings.

7. Layout

Grid

Mobile is single-column. Card-grids (floor map, KPI rows) use fixed column counts:

- **Floor map:** 4 columns on a 720p screen. For 8–10 tables it produces 2–3 rows; for 16–20 tables it produces 4–5 rows — both fit one viewport when paired with a compact pulse row above.
- **KPI tile row:** 3 columns for status pulse (Tables / Kitchen / Late). 4 columns for KPI mini-tiles (Orders / Open / Avg / Profit).
- **Quick actions:** 4 columns for the action grid.

Column gaps use `sp3` between siblings.

Container widths

Context	Max width
Mobile content	Full screen width minus `sp4` × 2
Tablet content (landscape)	720 px centered
Tablet content (portrait)	Full screen width minus `sp5` × 2, max 600 px
Modal sheets	Full screen width on mobile; 480 px centered on tablet
Onboarding / first-run	Max 480 px centered

On tablets, never let a single card stretch to 1024 px wide — analytics charts become unreadable. Cap at 720 px and add side gutters.

Aspect ratios for media

| Media | Ratio |

|---|---|
| Hero image (homepage, storefront) | 16:9 |
| Menu item photo | 4:3 |
| Restaurant logo | 1:1 (square crop) |
| Receipt thumbnail | 3:4 |
| AI scan capture preview | 4:3 |

All media must be cropped to one of these ratios. Native aspect ratio uploads are auto-cropped at the upload layer with a centered focal point, user-adjustable via the upload editor.

Safe area & system bars

- Every screen wraps its body in `SafeArea` with `top: true, bottom: false`.
- Bottom inset is handled by the bottom nav, which sits below the shift strip but above the system gesture inset.
- The status bar uses the `paper` color with dark icons by default. Inverted only on full-dark hero screens (rare).

Stacking order at screen bottom

From bottom of viewport upward, the documented stack is:

1. System navigation inset
2. `TfBottomNav` — always present on tabbed screens
3. `TfShiftStrip` — on Home/Manage dashboard only, sits directly above bottom nav
4. `TfStickyCTA` — on flow screens (order entry, forms) only, sits directly above bottom nav (mutually exclusive with shift strip)
5. `TfFab` — when present, anchored bottom-right with `sp4` margin from shift strip / bottom nav above

The FAB never overlaps a sticky CTA. On screens with a sticky CTA, the FAB is suppressed.

Edge gutter

Default screen edge padding is `sp4` (16 px). On tablet, increases to `sp5` (24 px) until container max-width is reached.

8. Shadow (`PosShadows`)

Helio G35 sets the ceiling. Tokens calibrated to scroll smoothly on PowerVR GE8320.

| Token | Spec | Use |

|---|---|---|
| `none` | empty list | Default. Cards rely on 0.5 px border. |
| `soft` | `0 1px 2px rgba(15,20,25,0.05)` | Subtle lift for stat tiles. |
| `glow` | `0 4px 12px rgba(15,20,25,0.06)` | Raised cards, FAB. |
| `raised` | `0 8px 24px rgba(15,20,25,0.08)` | Sticky CTAs, sheets, modals. |

Hard performance rules

- `blurRadius` ≤ 24 anywhere. Most surfaces ≤ 12 .
- `boxShadow` lists have exactly one entry.
- No `BackdropFilter`, no `ImageFilter.blur`.
- No `surfaceTintColor` overrides — theme sets `Colors.transparent`.
- Default for new cards is `none`. Use `soft`/`glow` only when elevation is meaningful.

12. Component catalog (`Tf\*`)

All in `tf\_design\_system.dart`. Reuse first; extend with optional parameters.

Core text & money

Component	Purpose
`TfText`	All text. Localizes digits and font.
`TfMoney`	Currency with ₵ and tabular figures.
`TfTextPair`	Locale-correct EN/BN string with fallback.

Navigation & chrome

Component	Purpose
`TfAppBar`	Screen header, rendered inside `Scaffold.body`.
`TfBottomNav`	Footer nav. Active item: accent icon, 2 px accent underline.
`TfTabs`	Linear segmented control. Active: ink background, surface text.
`TfShiftStrip`	Sticky dark strip above bottom nav on Manage dashboard.
`TfOfflineBanner`	Top-of-screen sync banner, warning tint.
`TfPrinterPip`	Global indicator of Bluetooth printer state. Sits in app bar trailing.

Buttons & actions

| Component | Purpose |

|---|---|

| `TfButton` | All buttons. Variants: `primary`, `dark`, `ghost`, `paper`. Sizes: `lg` (52), `md` (44), `sm` (32). |

| `TfIconButton` | 44×44 icon target with optional badge. `sm` variant (36×36) for dense toolbars. |

|

| `TfFab` | 56×56 accent FAB with `glow` shadow. |

| `TfStickyCTA` | Pinned bottom action with `raised` shadow. |

| `TfQuickAction` | Tile for the dashboard quick actions row — icon top, label bottom, 72 px tall. |

Surfaces

| Component | Purpose |

|---|---|

| `TfCard` | Bordered surface, `md` radius, `none` elevation by default. |

| `TfRail` | Coloured accent stripe in row tiles. |

Selection & status

| Component | Purpose |

|---|---|

| `TfChip` | Toggleable pill. Active: `accentSoft` background, `ink` text. |

| `TfStatusBadge` | Status pill. Kinds: `pending`, `accepted`, `late`, `served`, `info`, `warning`, `danger`. |

| `TfTableTile` | Floor-map tile. States: idle, seated, in_kitchen, served, late, bill_pending. Shows table number + state-colored fill + duration. |

Forms

| Component | Purpose |

|---|---|

| `TfField` | Form input. `sm` radius. |

| `TfSearchField` | Standalone search. 48 height, `sm` radius. |

| `TfNumPad` | Custom numeric keypad for inventory counts, cash entry, staff PINs. Bypasses the OS keyboard. |

Data display

| Component | Purpose |

|---|---|

| `TfDeltaBadge` | +/- percent with signal color (success up, danger down, muted flat). Brand accent never appears. |

| `TfMiniSparkline` | Inline chart for stat cards. Height ≤ 40 px. |

| `TfProportionalBar` | Horizontal bar split into accent + muted segments. |

| `TfGhostBar` | Side-by-side this-period vs last-period comparison bars. |
| `TfKpiTile` | Number + label + optional mini-sparkline + optional delta. The workhorse of
dashboards. |

Feedback

Component	Purpose
`TfEmptyState`	No data because nothing has happened yet. Icon + title + message + CTA.
`TfNoDataState`	Not enough data yet. Single muted explanation, no chart.
`TfLoading`	Async spinner block.
`TfSkeleton`	Shimmer placeholder for async content. Used on dashboards, lists, menu grids.
`TfToast`	Transient feedback ("Saved", "Sent to kitchen", "Try again"). Top-anchored,
3-second auto-dismiss.	
`TfConfirmSheet`	Bottom-sheet confirmation, fire-and-forget.

Section structure

Component	Purpose
`TfSectionHeader`	Uppercase 11 px eyebrow label.

Icons

Component	Purpose
`TfIcon`	All icons. Wraps Lucide with size and color discipline.

13. Patterns

Screen title

`TfAppBar` inside body, never `Scaffold.appBar`. Subtitle reports state, not chrome.

```
``dart
TfAppBar(
  title: 'Orders',
  subtitle: '36 in kitchen · 0 pending',
  trailing: [
    const HeaderLanguageButton(),
    const TfPrinterPip(),
    HeaderNotificationBell(),
```

```
],  
)  
...
```

Lists and grouped rows

```
``dart  
TfCard(  
  padding: EdgeInsets.zero,  
  child: Column(children: [  
    for (var i = 0; i < items.length; i++) ...[  
      _TileRow(item: items[i],  
        if (i < items.length - 1)  
          Divider(height: 1, color: PosColors.lineStrong),  
    ],  
  ]),  
)  
...
```

Floor map (8–20 tables)

```
``dart  
GridView.count(  
  crossAxisCount: 4,  
  shrinkWrap: true,  
  physics: const NeverScrollableScrollPhysics(),  
  mainAxisSpacing: PosSpacing.sp3,  
  crossAxisSpacing: PosSpacing.sp3,  
  children: tables.map((t) => TfTableTile(state: t)).toList(),  
)  
...
```

Forms

`TfField` for inputs. Numeric inputs (quantities, prices, counts) open `TfNumPad` instead of OS keyboard. Validation flows through `errorText`.

Destructive flows

```
``dart  
TfConfirmSheet.show(  
  context,  
  title: 'Cancel order #37?',  
  description: 'This cannot be undone. Kitchen will be notified.',
```



```
confirmLabel: 'Cancel order',
isDanger: true,
onConfirm: () async { ... },
);
...
```

Loading states

- Initial mount: `TfSkeleton` placeholder for each expected section.
- Background refresh: silent, no spinner — `TfToast` on completion if state changed.
- User-initiated action: `TfButton(busy: true)` inline.

Empty vs no-data

- `TfEmptyState` when nothing has happened yet ("No orders today — start your first").
- `TfNoDataState` when more time/data is needed ("Trends appear after 7 days of orders").

Delta display

```
```dart
TfDeltaBadge(value: -0.61, period: 'vs yesterday')
...
```

The widget enforces signal color from value sign. Impossible to render a negative delta in accent.

### ### Transient feedback

```
```dart
TfToast.success(context, 'Order #38 sent to kitchen');
TfToast.warning(context, 'Printer disconnected — retrying');
TfToast.error(context, 'Could not save — try again');
...
```

Toasts are top-anchored, 3-second dismiss, max one visible at a time.

Offline state

`TfOfflineBanner` above the body when `app.isOffline`. Locale-aware.

14. Voice and language

The voice doc is part of the system. Mismatched voice undermines consistent visuals.

We write

- Imperative button labels: "Send to kitchen", "Print receipt", "Close today"
- Plain operator language: "4 open", "Looks good", "Try again"
- Always the unit: "4.2 turns per day", "42 min avg", "৳652 per cover"
- BN strings hand-written, never machine-translated

We don't write

- "Submit order" → "Send to kitchen"
- "Generate document" → "Print receipt"
- "Validation successful" → "Looks good"
- "An error occurred" → "Try again"
- "4 pending orders requiring action" → "4 open"
- "৳652.00 avg" → "৳652 per cover"

Number formatting

- Currency rounds to whole taka unless reconciliation.
- Percentages: no decimal unless precision matters.
- Durations: "42 min", "1h 10m". Never raw float with "m".
- Times: 12-hour with am/pm in EN; Bengali equivalents in BN.

15. Multi-user discipline

8–20 tables and 5–6 waiters means handovers and attribution matter.

Attribution

- Every order is stamped with the creating staff member's account ID.
- Every modification (item added, item removed, bill split) is logged with timestamp + staff ID.
- The "Edited by Saif at 7:42 PM" footer appears on any modified order.

Shift discipline

- Shift open and close are explicit actions, not implicit.
- Each waiter has their own PIN-protected shift state.
- Closing a shift requires reconciling cash drawer; the app warns about open orders before allowing close.

- A waiter handing off a busy floor opens a "shift handover" sheet that lists open tables, open orders, and pending bills with explicit transfer to the next waiter.

Soft delete & recovery

- Cancelling an order moves it to a `cancelled` state, never deletes.
- Cancelled orders remain visible in the Orders list under a filter tab for 24 hours, then archive.
- A cancelled order can be restored within 10 minutes by the waiter or manager.

Concurrent edit

- If two devices try to modify the same order, the second device sees a "This order was just updated by [Name]" toast and refreshes.
- Last-write-wins is acceptable for non-critical fields (note, table number). Item-level changes use a server-resolved merge.
